

Ruffed Grouse Winter Habitat Use in Mixed Softwood–Hardwood Forests, Québec, Canada

PIERRE BLANCHETTE,¹ *Ministère des Ressources naturelles et de la Faune du Québec, Direction de la recherche sur la faune, 880, Chemin Sainte-Foy, 2^e étage, Québec, PQ G1S 4X4, Canada*

JEAN-CLAUDE BOURGEOIS, *Ministère des Ressources naturelles et de la Faune du Québec, Direction de l'aménagement de la faune, 5575, St-Joseph, Trois-Rivières-Ouest, PQ G8Z 4L7, Canada, and Université du Québec à Trois-Rivières, Département de chimie-biologie, C.P. 500, Trois-Rivières, PQ G9A 5H7, Canada*

SYLVAIN ST-ONGE, *Ministère des Ressources naturelles et de la Faune du Québec, Direction de la recherche sur la faune, 880, Chemin Sainte-Foy, 2^e étage, Québec, PQ G1S 4X4, Canada*

ABSTRACT Ruffed grouse (*Bonasa umbellus*) is a popular small game species in northeastern North America. We assessed female ruffed grouse habitat characteristics during winter of 2001–2002 and 2002–2003 in a region dominated by mixed softwood–hardwood forests by comparing used and random locations. We followed 23 radiotagged adult females in 2 forest sites of the Réserve faunique de Portneuf, Québec, Canada, from late November to mid-April. We described grouse habitat using ground surveys and identified selected habitat characteristics using analysis of variance and logistic regression. Females preferred mixed softwood–hardwood stands >17 m tall and 61–120 years old. Compared with random locations, grouse locations had more well-developed total canopy cover (>4 m; 75%), canopy (>4 m; 35%), midstory (1–4 m tall; 35%), and lower-story (<1 m tall; 23%) coniferous cover, and higher coniferous stem density and tree basal area (dbh > 9 cm; 343 stems/ha and 9.0 m²/ha, respectively). Forest management should maintain mature mixed softwood–hardwood stands (50% coniferous), which are adequate winter habitat for ruffed grouse. (JOURNAL OF WILDLIFE MANAGEMENT 71(6):1758–1764; 2007)

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Ruffed grouse (*Bonasa umbellus*) use different habitats during a year (Bump et al. 1947, Gullion and Svoboda 1972, Gullion 1977a). More particularly, adult females use young forests with dense regeneration in summer for brood rearing (Kubisiak 1978, Scott et al. 1998, Haulton et al. 2003, Giroux et al. 2007) and older forests the rest of the year (Berner and Gysel 1969, Thompson et al. 1987, Larson et al. 2003). Ruffed grouse winter habitat has been described in areas dominated by aspen (*Populus* spp.) forests where snow accumulation can be significant. In these conditions, grouse used aspen stands at the sapling or mature stage with low shrub density (Berner and Gysel 1969). During winter, grouse feed mostly on aspen buds or twigs (Doerr et al. 1974). Furthermore, ground snow is deeper in mature deciduous stands, which permits snow roosting, thereby minimizing energy loss due to thermoregulation (Brander 1965). Coniferous trees can be used as roost sites and thermal cover, but high tree pines (*Pinus* spp.) can also be detrimental to grouse as they offer more opportunities to predators (Gullion and Marshall 1968). Ruffed grouse winter habitat has also been studied in the central and southern parts of the species' range dominated by oak (*Quercus* spp.) forests where snow accumulation is not high enough to allow snow roosting. In these conditions, grouse used mixed stands of red cedar (*Juniperus virginiana*) and deciduous species (Thompson and Fritzell 1989). Grouse used sites offering thermal cover, such as coniferous stands, deciduous stands of species with persistent leaves during winter (e.g., oak), or branch piles (Thompson and Fritzell 1988a, Whitaker and Stauffer 2003).

In Québec, the highest relative abundances of ruffed grouse were observed in sugar maple (*Acer saccharum*)–yellow birch (*Betula alleghaniensis*)–American beech (*Fagus grandifolia*), balsam fir (*Abies balsamea*)–yellow birch, and balsam fir–white birch (*B. papyrifera*) bioclimatic domains (Doyon 1995). These domains are within a transitional zone between temperate deciduous forest and boreal coniferous forest. Mixed softwood–hardwood stands, and a juxtaposition of pure deciduous and pure coniferous stands, characterize this zone, and this landscape heterogeneity and complexity represent a constant challenge to managers (Girard et al. 2004). Aspen is present but does not dominate the landscape except in some areas.

The Québec Forest Act promotes sustainable forest development to meet multiple resource objectives such as producing wood fiber and managing wildlife habitat. Ruffed grouse is a popular small game species in Québec as it is elsewhere in northeastern North America (Bourret 1992, Drogin-Rodgers et al. 2003) and hunting does occur on public land open to commercial forestry activities. Managers of public territories need to concentrate on methods and strategies to improve or enhance ruffed grouse hunting to provide additional recreational user days and still maintain commercial forestry activities. Gullion and Svoboda (1972) developed a classic mosaic of 1-ha clearcuts for aspen forests, but it is questionable for use in modern forestry (Gullion 1977b, Kubisiak et al. 1980, Wiggers et al. 1992, McCaffery et al. 1996, Stoll et al. 1999) and does not improve in hardwood–softwood nordic-temperate forests. Our objective was to characterize winter habitat used by adult female ruffed grouse to provide information needed to manage ruffed grouse habitat for northern mixed softwood–hard-

¹ E-mail: pierre.blanchette@mrnf.gouv.qc.ca

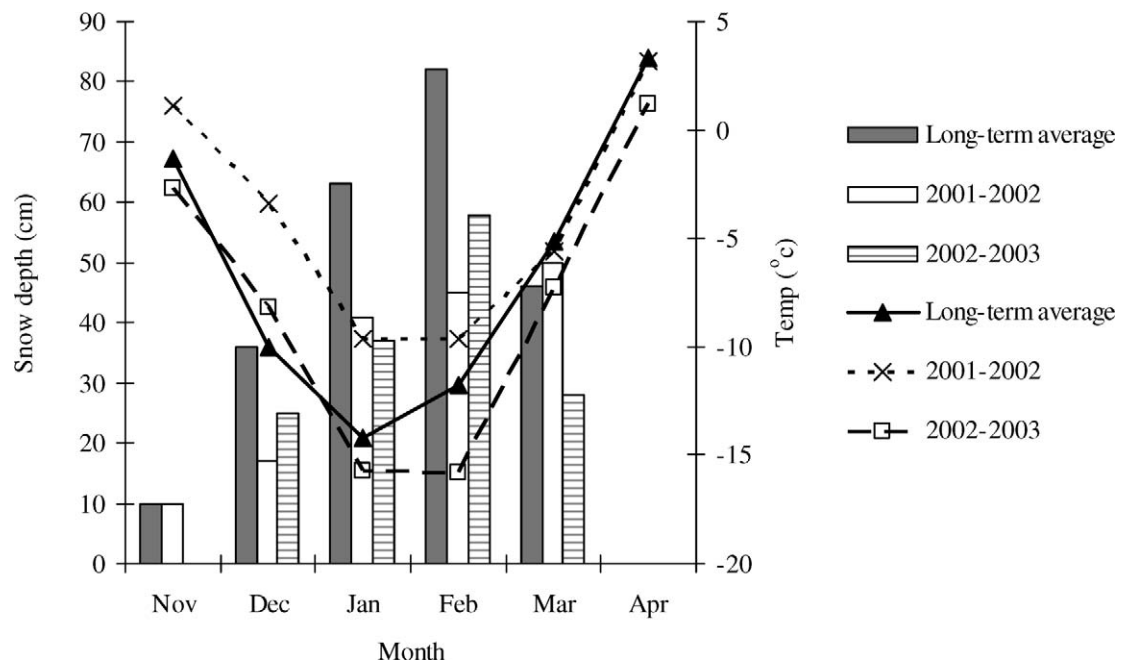


Figure 1. Mean daily temperature (lines) and snow depth at month-end (bars) by month for the period of 1971–2000 (long term) and winters of 2001–2002 and 2002–2003 at the station of Lac aux Sables, Québec, Canada (46°52'N, 72°24'W; Environment Canada 2004).

wood forests. We compared habitat use with availability in 2 forest settings (study sites) of the Réserve faunique de Portneuf, Québec, Canada, to identify important winter habitat characteristics. We hypothesized that female ruffed grouse would use deciduous stands at the sapling or mature stage because snow accumulation is abundant during winter.

STUDY AREA

We conducted our study in 2 forest settings (study sites)—Belles-de-Jour (BDJ: 23 km²) and Landry (LAN: 17 km²)—of the Réserve faunique de Portneuf, Québec, Canada (47°6'N, 72°9'W) during winter 2001–2002 and 2002–2003. The Réserve was located in the boreal hardwood transition region of the Bird Conservation Regions. In our study sites, 3 major vegetation zones were present: sugar maple–yellow birch, balsam fir–yellow birch, and balsam fir–white birch. Mean annual temperature was 2.5° C and mean annual precipitation was 1,000–1,600 mm, of which 30% was snow (Robitaille and Saucier 1998). Open areas, roads, trails, lakes, and streams were scattered throughout both study sites. Elevation ranged from 150 m to 650 m with gentle to steep slopes. Mean daily temperatures were slightly warmer during winter 2001–2002 and slightly colder during winter 2002–2003 than the long-term average (Fig. 1) measured at the nearest meteorological station to the study sites (35 km; Lac aux Sables, 46°52'N, 72°24'W; Environment Canada 2004). Snow depths at the month's end were below the long-term average for both winters, except for March of 2002, when snow depth was slightly above average (Fig. 1). Snow was powdered all winters long except in April of both years. No ice storm occurred during our study.

The 2 study sites differed mostly by stand cover type, age, height, and past forest management, based on ground

surveys of randomly distributed sample plots used to estimate available habitat (Table 1). A greater proportion of young stands composed LAN than composed BDJ. The tree layer in both areas was composed mostly of balsam fir, yellow birch, black spruce (*Picea mariana*), sugar maple, red maple (*A. rubrum*), and white birch. Trembling and large-

Table 1. Forest characteristics of study sites Belles-de-Jour (BDJ) and Landry (LAN) in the Réserve faunique de Portneuf, where we studied ruffed grouse in Québec, Canada, 2002.

Forest characteristics	% of random sites	
	BDJ (n = 74)	LAN (n = 73)
Stand age		
2–20 yr old	18	33
20–60 yr old	7	21
60–100 yr old	16	30
Uneven-aged stands	59	16
Stand ht		
1.5–7 m tall	23	40
7–18 m tall	28	37
>18 m tall	49	23
Cover type		
Deciduous	36	22
Mixed-deciduous	15	10
Mixed-coniferous	16	12
Coniferous	15	22
Regeneration	18	34
Commercial logging		
Uncut	47	47
Clearcuts 2–10 yr old	8	6
Clearcuts 11–20 yr old	10	28
Partial cuts (<20 yr old) ^a	35	19

^a Partial cuts include selection cutting, commercial thinning, improvement cutting, partial cutting, and shelterwood cutting.

toothed aspen (*P. tremuloides* and *P. grandidentata*) composed <1% of total basal area in the randomly distributed sample plots. The shrub layer in both sites was composed mostly of mountain maple (*A. spicatum*), mooseberry (*Viburnum alnifolium*), pin cherry (*Prunus pensylvanica*), beaked hazelnut (*Corylus cornuta*), ground hemlock (*Taxus canadensis*), berry species (*Rubus* spp.), and the previously mentioned tree species. Grouse population density was relatively low as expressed by the historical fall hunting success of the reserve, which was at its lowest in the last 9 years (P. Blanchette, Ministère des Ressources naturelles et de la Faune du Québec, unpublished data).

METHODS

Capture and Telemetry

Searchers captured female grouse on roads or trails during June and July of 2001 and 2002. Searching effort was similar in both study sites. We sampled all stand types available to ruffed grouse because roads and trails were scattered throughout both study sites except in inaccessible terrain (slope >40%, swamps, marshes, etc.). We first lured female grouse using a recorded distress call of a lost chick and then captured them with a fishing net or snare perch (Lyons 1981). We marked adult female grouse with 2 numbered plastic leg bands and fitted each bird with a R1-2BM Holohil transmitter (Holohil Systems Ltd., Carp, ON) using a backpack kevlar harness (15–17 g; Brander 1968). We released birds at their capture site within 10 minutes of capture. The Québec Faune et Parcs committee on animal care approved all bird handling techniques (certificate no. 00–01).

We located radiomarked grouse by homing once per week from late November to mid-April of 2001–2002 and 2002–2003. We randomly located each grouse from 0830 hours to 1530 hours because ruffed grouse are considered inactive at night (Maxson 1977, Pietz and Tester 1982). We used a handheld Telonics antenna and an ATS R2100 (Advanced Telemetry Systems Inc., Isanti, MN) receiver to locate grouse. We took care not to scare off individuals to minimize disturbance and accurately locate females. We recorded geographical coordinates using a Garmin XL Global Positioning System (GPS; 15-m accuracy) and attached an identified flag to the nearest tree for later habitat measurements.

Habitat Measurements

We measured most habitat characteristics in May 2002 and 2003, before leaf growth, at grouse locations and at random locations. However, we measured lateral obstruction, midstory and lower-story cover, small stem density, and snow depth during winter, when we located grouse, because these variables are influenced by snow accumulation. We generated random locations within both study sites using ArcView 3.2 Geographic Information System (GIS). We transferred geographical coordinates of random locations to a Garmin XL GPS for location in the field. We spaced random locations >50 m from each other and rejected those found on roads, water bodies, and inaccessible areas. There

was no overlap between random and known grouse locations. We did not take habitat measurements in grouse locations that were <20 m from a previous grouse location to avoid pseudoreplication. We used a total of 147 random locations to determine habitat availability (73 in BDJ and 74 in LAN) and measured variables influenced by snow accumulation in 126 of these locations (68 in BDJ and 58 in LAN). We adapted habitat measurements from Dunn and Braun (1986), Thompson et al. (1987), and Scott et al. (1998).

We recorded stand type (deciduous: <25% of coniferous basal area [BA]; mixed-deciduous: 26–50% coniferous BA; mixed-coniferous: 51–75% coniferous BA; coniferous: >76% coniferous BA), stand height, and presence of commercial logging (clearcuts or partial cuts) within a 0.04-ha circular plot centered on each grouse location. We considered selection cutting, commercial thinning, improvement cutting, and shelterwood cutting partial cuts. The observer estimated the slope position of the plot as lowland, down slope, midslope, upper slope, or summit. We determined stand age, commercial logging year, and distance to nearest clearcut and distance to nearest road from 2001 digitized topographic-forest maps of the Ministère des Ressources naturelles et de la Faune du Québec (scale 1:20,000) produced using ArcView GIS 3.2 software.

We estimated canopy cover (>4 m tall), midstory cover (1–4 m tall), and lower-story cover (<1 m tall) for deciduous species, coniferous species, and all species (total). We estimated cover using the point intercept method at 3-m intervals along a north-south 30-m transect centered on the location point (Gysel and Lyon 1980). We assessed lateral obstruction using a 0.5 × 2.0-m density board divided into 4 0.5 × 0.5-m squares and placed 15 m from the standing observer (Nudds 1977). We took lateral obstruction measurements in each cardinal direction while standing at the centre of each 0.04-ha plot. We determined diameter at breast height and tree density (≥9 cm dbh) for coniferous, deciduous, and all tree species by total stem counts within the 0.04-ha circular plot. We measured density of small woody species (<9 cm dbh and >50 cm tall), including shrubs, in 5 different locations, which were the center and each cardinal direction of the 0.04-ha plot perimeter (11.28 m from the center). We measured snow depth (±0.5 cm) at 10 locations uniformly scattered in the 0.04-ha plot.

Data Analysis

We derived 29 habitat variables from the habitat measurements. We pooled data from the 2 years of study to take into account temporal heterogeneity. We analyzed categorical variables using a log-linear model generalized for the Poisson distribution to examine whether grouse selection patterns differed between study sites (Heisey 1985). When log-linear models indicated no significant difference between study sites, we analyzed the variable by combining data from both sites. Otherwise, we analyzed the variable for sites separately. We compared habitat use versus availability

Table 2. Habitat characteristics of ruffed grouse locations in winter ($n = 262$) versus random locations ($n = 147$) in the Réserve faunique de Portneuf, Québec, Canada, winter of 2001–2002 and 2002–2003.

Habitat variable	% use	% available	95% CI	Use vs. availability ^a
Stand type				
Regeneration	11.5	26.5	0.232 to 0.070	–
Deciduous	17.2	29.3	0.207 to 0.034	–
Mixed-deciduous	26.0	12.2	–0.062 to –0.212	+
Mixed-coniferous	32.8	14.3	–0.105 to –0.266	+
Coniferous	12.6	17.7	0.125 to –0.023	=
Stand age				
2–20 yr old	11.2	25.3	0.222 to 0.062	–
21–60 yr old	13.1	19.9	0.144 to –0.009	=
61–120 yr old	45.6	30.8	–0.051 to –0.244	+
Uneven-aged stands	30.1	24.0	0.027 to –0.150	=
Stand ht				
1.5–4 m	6.1	31.3	0.332 to 0.171	–
5–7 m	32.4	25.9	0.025 to –0.157	=
8–12 m	9.9	6.8	0.023 to –0.086	=
>17 m	51.5	36.1	–0.056 to –0.253	+
Commercial logging				
Clearcuts 2–10 yr old	0.4	6.8	0.106 to 0.023	–
Clearcuts 11–20 yr old	11.1	18.5	0.148 to 0.001	–
Partial cuts	25.2	27.4	0.111 to –0.067	=
Uncut	63.4	47.3	–0.061 to –0.261	+
Slope position				
Lowland	17.9	22.4	0.127 to –0.037	=
Down slope	14.5	21.8	0.152 to –0.007	=
Midslope	46.9	31.3	–0.060 to –0.253	+
Upper slope	7.6	12.9	0.116 to –0.010	=
Summit	13.0	11.6	0.052 to –0.080	=

^a (–), (+), and (=) indicate habitat categories used in less, greater, or equal proportion to availability, respectively.

using a chi-square test for homogeneity and simultaneous 95% confidence intervals (Marcum and Loftsgaarden 1980, Jelinski 1991).

We analyzed each continuous variable using 2-way analysis of variance (ANOVA). The factors for each analysis were site (BDJ vs. LAN) and habitat use (telemetry vs. random locations). When the study site factor significantly affected the use-availability relationship, we made multiple comparisons separately for both areas using a least-square means test. We transformed data to tend toward normality and homogeneity of variances, but ANOVA is robust to departure of these assumptions when a large sample size is used (Hair et al. 1987).

We used a stepwise logistic regression analysis to generate a ruffed grouse habitat model for both study areas combined. We retained all variables with $P < 0.15$ in previous univariate analyses for investigation (Hosmer and Lemeshow 2000). The stepwise logistic regression model reduces model overspecification and multicollinearity by allowing elimination of variables that no longer increase model performance (Hosmer and Lemeshow 2000). We used the Wald chi-square test to evaluate the importance of variables

selected in the regression, a larger chi-square value indicating higher relative importance. We used the area under the Receiver Operating Characteristic (ROC) curve to assess the fit of the model (Hosmer and Lemeshow 2000). We also used the Hosmer–Lemeshow statistic (HLs) to assess goodness-of-fit of the model. A P value >0.05 indicated that frequencies of observed and predicted values of the model were not significantly different (Hosmer and Lemeshow 2000).

Logistical constraints prevented the use of additional locations to test our model. We therefore calculated the percentage of sites classified correctly (relocated vs. random) by the logistic regression model using the leave-one-out cross-validation method (Stone 1977). We repeated this procedure for each of the 377 observations used to build the model. This pseudovalidation procedure classified each observation using the logistic function developed from all observations except the one being classified. We considered sites with a predictive probability of ≥ 0.68 (no. grouse occurrences/no. total observations) to predict the presence of grouse (Hosmer and Lemeshow 2000). We performed statistical analyses using SAS/STAT 8.02 (SAS Institute 1999). We considered results significant when $P < 0.05$.

RESULTS

We followed 23 different adult female grouse, 12 in BDJ and 11 in LAN. We determined habitat preference using 10.1 ± 0.8 ($\bar{x} \pm \text{SD}$) relocations per grouse for a total of 262 grouse locations (133 BDJ, 129 LAN). Female ruffed grouse used mixed stands and avoided deciduous and regeneration stands ($\chi^2 = 41.6$, $\text{df} = 4$, $P < 0.001$; Table 2). They used stands 61–120 years old and avoided stands 2–20 years old ($\chi^2 = 20.5$, $\text{df} = 3$, $P < 0.001$). They used stands ≥ 17 m tall and avoided stands 1.5–4 m tall ($\chi^2 = 46.9$, $\text{df} = 4$, $P < 0.001$). They avoided clearcuts < 20 years old and used uncut mature stands (> 21 yr old; $\chi^2 = 22.7$, $\text{df} = 3$, $P < 0.001$). We found grouse locations mostly in midslope ($\chi^2 = 18.2$, $\text{df} = 4$, $P = 0.001$).

Grouse locations were characterized by a higher proportion of coniferous cover than random locations for canopy (75.0% vs. 65.2%; $P \leq 0.01$), midstory (35.4% vs. 27.0%; $P \leq 0.01$) and lower story cover (22.7% vs. 16.8%; $P < 0.05$; Table 3). Habitat use by female grouse was not related to lateral obstruction (Table 3). Coniferous large stem (dbh ≤ 9 cm) BA and stem density were higher at grouse locations than at random locations (BA = 9.0 m²/ha and 6.3 m²/ha; stem density = 342.8 stems/ha and 300.7 stems/ha, respectively, $P \leq 0.01$). Total density of small woody stems (≥ 50 cm tall and dbh < 9 cm) was lower at grouse locations (15,727 stems/ha) than at random locations (19,329 stems/ha).

Grouse locations were closer to roads than random locations in BDJ but were farther from them in LAN. Grouse locations were closer to clearcuts 2–20 years old than random locations in BDJ only (Table 3). Snow depth was 7.5 cm higher at random locations (56.9 cm) than at grouse locations (49.4 cm).

Table 3. Habitat characteristics of female ruffed grouse locations in winter versus random locations in the Réserve faunique de Portneuf, Québec, Canada, winter of 2001–2002 and 2002–2003.

Habitat variables	Combined sites				Belles-de-Jour (BDJ)				Landry (LAN)			
	Used		Random		Used		Random		Used		Random	
	(n = 262)		(n = 147)		(n = 133)		(n = 74)		(n = 129)		(n = 73)	
	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE
Cover (%)												
Total canopy closure (>4 m)	75.0	2.1	65.2**	2.9								
Deciduous canopy closure					53.8	2.6	60.3	3.0	52.1	2.5	33.9**	3.2
Coniferous canopy closure	34.6	2.3	23.4**	2.6								
Total midstory cover (1–4 m)	68.7	1.9	68.9	2.5								
Middeciduous cover	43.3	2.4	51.7	2.9								
Midconiferous cover	35.4	2.1	27.0**	2.3								
Total lower story cover (<1 m)	40.8	2.1	49.9*	2.4								
Lower deciduous cover	20.0	1.7	32.9**	2.4								
Lower coniferous cover	22.7	1.9	16.8*	1.8								
Lateral obstruction (%)												
0–2 m tall	47.8	0.9	44.4	1.5								
0–1 m tall	52.4	1.2	47.1	1.5								
1–2 m tall	42.3	1.2	39.4	1.5								
Large stems (dbh ≥ 9 cm)												
Basal area (m ² /ha)												
Total					18.0	0.7	15.8**	1.2	20.1	0.8	14.0**	1.3
Deciduous stems					9.6	0.5	10.4	0.9	10.4	0.6	6.8	1.0
Coniferous stems	9.0	0.5	6.3**	0.7								
Density (stems/ha)												
Total	621.7	21.4	512.4**	33.4								
Deciduous stems					292.5	14.9	278.4	22.5	264.7	14.5	144.2**	18.5
Coniferous stems	342.8	20.4	300.7**	33.8								
Small stems (≥50 cm tall and dbh <9 cm)												
Density (stems/ha)												
Total	15,727	675	19,329**	1,190								
Deciduous	13,453	648	16,103	1,156								
Coniferous					2,220	292	2,448	451	2,235	394	4,016**	563
Distance to edge (m)												
Road					93.8	5.5	212.0**	17.8	218.5	15.8	154.3**	14.4
Clearcut 2–20 yr old					477.7	39.0	387.0**	63.7	144.7	11.7	194.0	26.3
Snow depth (cm)	49.4	1.1	56.9**	1.1								

* $P < 0.05$.** $P < 0.01$.

The stepwise logistic regression analysis produced a model using 9 variables to predict the presence of female ruffed grouse in winter (ROC curve = 0.81, HLs $\chi^2 = 6.908$, df = 8, $P = 0.547$; Table 4). The probability of ruffed grouse presence was positively related with stands with higher coniferous canopy closure, lateral obstruction, BA of coniferous stems, density of deciduous stems, and density of total small stems. The probability of ruffed grouse presence was negatively related with stands with higher middeciduous and lower-deciduous cover and density of total stems and small coniferous stems. We correctly classified 71.1% of grouse locations and 73.1% of random locations as a result of leave-one-out cross-validation method procedures.

DISCUSSION

Choice of winter habitat is thus influenced by availability of food sources and presence of adequate snow roosting sites or adequate coniferous perches to minimize energy losses (Thomas et al. 1975; Thompson and Fritzell 1988a, b). In forests dominated by aspen, grouse used aspen stands at the

sapling or mature stage with low shrub density (Berner and Gysel 1969). Aspen used for winter feeding were 16–25 years old (Gullion and Svoboda 1972). In Alberta, Rusch and Keith (1971) found that grouse used pure aspen stands less frequently in winter than in other seasons, but aspen remains an important component of winter habitat because it is the main food source (Doerr et al. 1974). In the absence of aspen, grouse feed on buds and twigs of deciduous shrubs and trees, mostly beaked hazelnut, service-berries (*Ame-lanchier* spp.), birches (*Betula* spp.), alder (*Alnus rugosa*), willows (*Salix* spp.), cherries (*Prunus* spp.), hawthornes (*Crataegus* spp.), etc. (Doerr et al. 1974, Woehr 1974, Vézina 1975, Gullion 1984, Servello and Kirkpatrick 1987). In our study, aspen represented <1% of deciduous total basal area in grouse locations, whereas yellow birch and sugar maple dominated deciduous total basal area. However, we believe our study did not take into account locations used during winter feeding aggregations, which take place at dawn and at dusk, because telemetric locations took place between 0830–1530 hours (Doerr et al. 1974).

In mixed stands, coniferous trees seemed to be an

Table 4. Results of the stepwise logistic regression model predicting female ruffed grouse presence versus random locations in winter in the Réserve faunique de Portneuf, Québec, Canada, winter of 2001–2002 and 2002–2003.^a

Variable (unit)	Estimate	SE	P value	Wald χ^2
Intercept	0.1446	0.6408	0.821	0.051
Density of deciduous stems (no./ha)	0.0080	0.0013	<0.001	36.990
Lower-deciduous cover (%)	−0.6415	0.1291	<0.001	24.701
Density of total stems (no./ha)	−0.0044	0.0009	<0.001	22.419
Density of coniferous small stems (no./ha)	−0.0001	0.00003	0.001	10.431
Lateral obstruction 0–1 m tall (%)	0.3477	0.1134	0.002	9.399
Density of total small stems (no./ha)	0.00004	0.00001	0.004	8.434
Basal area of coniferous stems (m ² /ha)	0.1150	0.0435	0.008	6.995
Middeciduous cover (%)	−0.2505	0.1086	0.017	5.701
Coniferous canopy closure (%)	0.3379	0.1691	0.046	3.992

^a Receiving Operating Characteristic curve = 0.81, Hosmer–Lemeshow statistics $\chi^2 = 6.91$, df = 8, $P = 0.547$.

important component of grouse winter habitat in our study area. Grouse preferred locations with a higher tree cover, tree stem density, and coniferous basal area compared with random locations. Environmental conditions in mixed stands prevented snow accumulation compared with deciduous stands and as a result, snow depth was lower in grouse locations than in random locations. However, snow depth in grouse locations was sufficient ($\bar{x} = 49$ cm) and snow was of good enough quality (except in Apr, during which it was crusted) to allow snow roosting, which is possible starting between an estimated 20 cm and 30 cm (Gullion 1967, 1970; Thompson and Fritzell 1988a). Observers noted roost positions for 247 of the 262 grouse locations. Nearly 39% of grouse were perched in conifers whereas 37% were snow roosted. Although we did not measure this parameter, conifer distribution in mixed stands is probably important in the choice of tree roosting sites. Conifers distributed in groves of a few to many trees might have greater insulation capacity than lone trees.

The logistic regression model predicted that grouse presence increases with coniferous components. In Michigan, Hammill and Moran (1986) considered conifers to be an optimal component of winter habitat because they provide grouse with protection and cover. Conifers are particularly important as thermal cover for perched grouse (Thompson and Fritzell 1988a). Balsam fir and black spruce, dominant coniferous species in our study area, have low branches that can offer cover for grouse moving on the snow (Woehr 1974 *in* Cade and Sousa 1985). However, when conifers, particularly pines, are too abundant, they can have a negative effect on grouse by offering more opportunities for terrestrial and avian predators (Gullion and Marshall 1968). In the logistic regression model, probability of grouse presence also increases with the number of deciduous tree stems and deciduous small stems, which may indicate that main food sources in winter (buds and twigs of deciduous trees and shrubs) and cover against raptor predation are important components in the choice of winter habitat. Thus, mixed stands seem to offer an optimal juxtaposition of roosting cover and food sources for ruffed grouse during winter.

MANAGEMENT IMPLICATIONS

We found that important habitat characteristics for ruffed grouse during winter are sites with mature mixed softwood–hardwood forest stands with high midstory cover, high tree stem density, and a basal area composed of 50% conifers. We also found that ruffed grouse avoided clear cuts <20 years old. To promote annual ruffed grouse habitat, the silvicultural system used should maintain mature mixed forest stands within a mosaic composed of habitat types essential for other seasonal needs like brood-rearing habitat (Stoll et al. 1999, Giroux et al. 2007). Cutting types and rotation time of the forest mosaic favorable to ruffed grouse will depend on tree species present, their growth rate, and their regeneration mode, which can vary with edaphic conditions and climate in the region.

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